SMC-BVS: method, evidence, theory, and next steps

Briefing for the advisor meeting · Felipe Bedoya · November 2026

Status words. [proved, checked]: proof written and independently checked line by line. [proved, sign-off pending]: proof written and independently checked; final sign-off on the statement not yet recorded. [proved, unchecked]: proof written, independent check not yet done. [in progress], [open]: as stated. All theorems are asymptotic in n, with $p = p_n \le n^A$ for an arbitrary fixed A (so $p \gg n$ is included) unless said otherwise; conditions are on (X, θ^*) only; named design families appear only as examples.

1 The method

Target. Gaussian linear model on standardised data with a g/ridge prior $\beta_S \mid \sigma^2 \sim N(0, n\sigma^2(X_S^\top X_S + I)^{-1})$ ($g = n$), $\sigma^2 \sim \mathrm{IG}(0.01, 0.01)$, and a Beta-Binomial$(1, p)$ size prior truncated at $K = \min(p, \lfloor n/2 \rfloor)$; the score of a model is $F(S) = \log \varpi(S) + \log m(S)$ and $\pi(S \mid y) \propto e^{F(S)}$ on $\Omega = \{|S| \le K\}$. The sampler is a sequential Monte Carlo sampler over model space that starts from an *exactly sampled* block-product law and tempers towards π:

1. **Blocks and exact start law π_0.** Cluster columns (complete linkage on $1 - |r_{jl}|$) into blocks of size $\le B$ ($B \le 16$); score every sub-model of every block against the full y; $\pi_0(S) \propto \varpi(S) \prod_b m(S \cap C_b)/m(\varnothing)$. Its normaliser $\bar{Z}_0$ is computed exactly (convolution of block size profiles) and π_0 is sampled exactly and iid.
2. **Bridge.** $\gamma_\varphi(S) = \gamma_0(S)\, e^{\varphi d(S)}$, $d(S) = \log \mathrm{BF}(S) - \sum_b \log r_b(S \cap C_b)$, $\varphi : 0 \to 1$; $\gamma_1 \propto \pi$.
3. **Grid calibration.** A pilot SMC chooses temperatures by population ESS ≥ 0.90; the final grid is rebuilt from the scored models with a restricted-ESS criterion (escalation tier if diagnostics fail).
4. **Islands.** R independent islands ($R = 4$ in the study), each: reweight by $e^{(\varphi_t - \varphi_{t-1})d}$, resample (multinomial, every stage), move (a round of p Gibbs coordinate updates, swaps, and one independence proposal from π_0; each step γ_φ-reversible). Evidence $\hat{Z}_r = m(\varnothing)\, \bar{Z}_0 \prod_t N^{-1} \sum_i e^{(\varphi_t - \varphi_{t-1}) d(S_i^{(t-1)})}$.
5. **Outputs.** $\hat{Z}$ (mean of $\hat{Z}_r$); Rao–Blackwell PIPs (conditional inclusion probabilities recorded before each Gibbs draw, averaged over islands); MPM $= \{\mathrm{PIP} > \frac{1}{2}\}$; **HPM = best model scored anywhere** (a ledger of every model scored: tables, pilot, islands, independence proposals, searches).
6. **Final local search with certificate.** A greedy search (also run from the mode of π_0) ends with a flag `local_certified`: no single deletion, addition or swap of the reported model scores higher.

What is unusual. (i) SMC over model space with an *exact* start law, so the bridge starts at a law we can sample and normalise exactly; (ii) each island's $\hat{Z}_r$ is unbiased for Z (Thm S1 below); (iii) the reported model is a *search result* over everything scored, returned with a local-optimality certificate, and the theory is about that reported object, not only about the posterior.

2 Empirical evidence (simulation study, $p \gg n$)

Design. $n = 200$, $p \in \{500, 1000\}$; six covariance designs S1–S6 (independent; AR(1) 0.5; equicorrelation 0.7 and 0.85 within groups of 5; groups plus a global factor; heterogeneous groups); $|S_0| = 14$ true variables with $|\beta| \in [0.8, 1.8]$; two signal scales $c \in \{1, 0.5\}$; 100 replicates per cell: $2 \times 6 \times 2 \times 100 = 2{,}400$ fits. Recovery means the selected set equals S_0 exactly; the 6 fits that hit the run-time limit count as misses.

	$c = 1$ (strong)		$c = 0.5$ (weak)	
Exact recovery, out of 600	$p = 500$	$p = 1000$	$p = 500$	$p = 1000$
SMC-BVS, HPM	**559** (93.2%)	**560** (93.3%)	**370** (61.7%)	**340** (56.7%)
SMC-BVS, MPM (RB PIP > 0.5)	559	561	338	321
SuSiE	585	582	220	205
Spike-and-slab lasso	569	582	303	255
Horseshoe (95% CI)	574	575	185	136
Adaptive lasso (EBIC, $k \le 37$)	493	512	253	197
MCP (pre-registered EBIC rule)	377	235	240	192
ABESS (pre-registered EBIC rule)	365	323	256	214
Lasso / elastic net (EBIC, $k \le 37$)	111 / 61	89 / 50	24 / 14	15 / 10

- **Weak signal ($c = 0.5$):** SMC-BVS is first or tied-first in 11 of 12 design cells and recovers 67 ($p = 500$) and 85 ($p = 1000$) more data sets than the best competitor (spike-and-slab lasso).
- **Strong signal ($c = 1$):** 4th of 9, behind SuSiE, spike-and-slab lasso and the horseshoe (about 93% vs 95–97%); almost every miss adds variables to S_0. It beats the lasso family, MCP and ABESS (their EBIC rules with $k \le 37$) in all 24 cells.
- **Where the misses come from.** In every one of the 565 completed misses the reported model has a strictly higher posterior score than S_0 (smallest gap 0.0049 nats): the error is the posterior's, not the search's; 0 detected search misses.
- **Against the exact posterior** (companion study, $n = 200$, $p \in \{20, 30\}$, 2,400 cases, full enumeration): SMC HPM =

exact HPM in 2,400/2,400 primary and 12,000/12,000 repeat fits; $\log \hat{Z} - \log Z$ has bias -0.0002, RMSE 0.012; mean per-fit max $|\text{PIP} - \text{PIP}^{\text{exact}}|$ is 0.0005 (RB).

- **Cost.** Median run time per fit 565–607 s at $p = 500$ and 1,550–1,616 s at $p = 1000$; 906 core-hours for all 2,400 fits; median run time grows by about $2^{1.4}$ when p doubles.

3 Theory: headline results

Notation: $\delta_n = 1 - \pi(S_0 \mid y)$, $r_n = \sqrt{\log n / n}$, $s_0 = |S_0|$ fixed, $\hat{S}$ = reported model, $\mathcal{O}$ = the run completes with output. Song & Liang (2015, JRSS-B) = SL below; their theorems are about the posterior of the regression density (Hellinger), with p up to $\exp(Cn^\alpha)$.

A. Posterior concentration and selection

Thm P1 (concentration). If $\|X\theta^*\|^2 \le Mn$, sparse-eigenvalue floors hold on the sets $S_0 \cup J$ with $|J| \le j_0$ (fixed) and $|J| \le m_n^\sharp = O(\log n)$, and beta-min $\theta^{*2}_{\min} \ge c\,\sigma^2 \log n/(n\phi_0^2)$ ($c > c_0$), then $\delta_n = O_P(r_n)$; exact HPM = MPM = S_0 w.p. $\to 1$; $\max_j |\text{PIP}_j - \mathbf{1}\{j \in S_0\}| \le \delta_n$; without beta-min, no mass on strict supersets of S_0. Random designs via an interface (e.g. Gaussian rows with a sparse eigenvalue floor; pervasive factors allowed for this result). [proved, checked]

Design-free size tail. With no eigenvalue condition: models larger than $m_n^\sharp = O(\log n)$ carry mass $O_P(n^{-1/2})$, uniformly over all X with $\|X\theta^*\|^2 \le nM$. [proved, checked]

Rate lower bound. For every design and data set with $1 \le s_0 < p$, $s_0 + 1 \le K$: $\delta_n \ge 1/(1 + \sqrt{n+1})$; so r_n is sharp up to $\sqrt{\log n}$. [proved, checked]

Strengthenings. (a) The eigenvalue floors away from S_0 can be replaced by an identifiability margin (every underfit S with $|S \setminus S_0| \le j_0$ leaves residual mean energy $\ge c\sigma^2 \log n$) plus a determinant-type (Occam-sum) condition on supersets: no eigenvalue condition on any set other than S_0. (b) $E\,\delta_n \le Cr_n$ uniformly over designs. (c) In-sample Hellinger contraction $\bar{h}_n^2 = O(\log n/n)$ without beta-min. [proved, sign-off pending]

Limit of selection. Specificity without beta-min fails: a numerical counterexample (null proxy with PIP $\to 1$). [counterexample, numerical; analytic sketch unchecked]

vs. Song–Liang: their Thm 1–2 give Hellinger consistency with exponential rates for p up to $\exp(Cn^\alpha)$; ours is model-level concentration with a rate and a matching lower bound, for $p \le n^A$, plus their in-sample analog (c). Their Thm 3 (sure screening) controls sensitivity only; ours gives sensitivity and specificity uniformly in j and exact HPM/MPM recovery. Our cap $K = \lfloor n/2 \rfloor$ violates their condition (10) when $p \ge n/2$; the size-tail lemma replaces it.

B. Misspecification and the start law

Thm M1 (misspecified candidate set). Run the method on a candidate subset $\mathcal{C}$, $|\mathcal{C}| \le n^A$, with omitted variables. If the omitted part is Gaussian given $X_{\mathcal{C}}$ (condition GR), the candidate posterior concentrates on the support $S_0^{\mathcal{C}}$ of the projection, at rate r_n; boundary examples show GR's content is needed. [proved, checked]
vs. Song–Liang: Thm 4 gives a.s. Hellinger consistency at the KL projection under their Condition 4; ours is exact-model concentration with a rate, under the stronger GR.

Start law. The coded sampler draws π_0 exactly and computes $\bar{Z}_0$ exactly [proved, unchecked; matched to code]. Under visibility/leakage conditions every mode $\tilde{S}$ of π_0 satisfies $S_0 \subseteq \tilde{S} \subseteq Q$, $|Q| = O(\log n)$ [proved, checked]. A rate $\pi_0(S_0) \ge 1 - n^{-c}$ in no-leakage designs [open; next].

C. The reported model

Thm C1 (certified screening; design-free). Fixed design, Gaussian errors; no condition on the start, blocks, leakage or calibration. If the sparse visibility $\Gamma_n(q)$ (noiseless gain of the best missing truth, minimised over models of size $\le q$ that miss a truth) satisfies $\sqrt{\Gamma_n(q_n)} \ge (1+\delta)[\tau_n\sqrt{2\log p + \log n} + \sigma\sqrt{2q_n \log p}]$, $\tau_n^2 = \sigma^2 + \|X\theta^*\|^2/(n-1)$, then every locally optimal model of size $\le q$ contains S_0; hence $P(\mathcal{O} \cap \{\text{certified}\} \cap \{|\hat{S}| \le q_n\} \cap \{S_0 \not\subseteq \hat{S}\}) \to 0$. Sufficient: a sparse eigenvalue floor on sets of size $q_n + s_0$ with $(q_n + 1)\log p + \log n \le c_1 n$. With s_0 bounded and a near-orthogonality floor $\phi \ge 2A/(2A+1) + \varepsilon$, a sparse certified model *equals* S_0. The constant 1 in the threshold cannot be lowered. [proved, checked]

Thm C2 (reported model under visibility and bounded leakage). Fixed design, Thm P1's conditions, eigenvalue floors on all sets of size ≤ 16, and a deterministic $Q \supseteq S_0$ with $|Q| \le (\log n)^\kappa$. Write $\delta^2_{U,l} = \|(P_{U\cup\{l\}} - P_U)X\theta^*\|^2$ (gain in explained mean when l joins U). (V) $\delta^2_{U,j} \ge M_n \log n$, $M_n \to \infty$, for every truth j and every $U \not\ni j$, $|U| \le 15$ (no truth is masked by ≤ 15 variables); (L) $\delta^2_{U,l} \le \eta\,\tau_U^2 \log n$ for every $l \notin Q$ and such U (τ_U^2 = residual variance left by U); (N) a lattice condition, automatic when $|Q \setminus S_0| \le (1 - \delta_0)\log n$. Then $P(\mathcal{O} \cap \{\hat{S} \ne S_0\}) \to 0$ for every seed, number of particles and islands, and calibration outcome. Gaussian-row version with a localisation clause. Examples: metric-local covariances at any $p \le n^A$ (including $p \gg n$), AR(1).

[proved, sign-off pending]; dropping (N) [open].

vs. Song–Liang: their MAP-model result is in an online supplement we could not obtain; their theorems concern the posterior, not the output of an algorithm. Their Condition 5 ($|\mathrm{cov}(x_t, y)| > 0$, which keeps true predictors in their screened set) is a marginal analogue of our conditional visibility (V).

D. Sampler outputs (the method as coded)

Thm S1 (unbiasedness). For every island, every measured round and every bounded h: $E[\hat{Z}_r\, N^{-1} \sum_i h(S_i) \mid \text{calibration}] = Z\, \pi(h)$; in particular $E[\hat{Z}_r \mid \text{calibration}] = Z$. No condition on X. [proved, checked]

Thm S2 (draw discovery). $P(\text{no draw of an island equals } S_0) = (1-\pi_0(S_0))^{D_r}$ exactly ($D_r =$ draws per island), so $P(\mathcal{O} \cap \{\hat{S} \neq S_0\}) \le P(\text{exact HPM} \neq S_0) + E(1-\pi_0(S_0))^{D_r}$; it suffices that $\max(n,p)\log_2(n+1)\,\pi_0(S_0) \to \infty$, with no condition on mixing or the bridge.

Thm S3 (concentrated bridge). For any grid and particle number, $E|\hat{Z}_r/Z - 1| \le 2\bar{\delta}$, $\bar{\delta} =$ the larger of the off-S_0 odds under π and π_0; with concentration of π and π_0: $\hat{Z}/Z \to 1$ and consistent PIPs, MPM and HPM.

Thm S4 (terminal repair). The $\varphi = 1$ move kernel is minorised by π/M, $M = \max \pi/\pi_0$; with posterior concentration and $L/M \to \infty$ ($L =$ burn-in rounds) the PIPs and MPM are consistent with no condition on $\hat{Z}$, the bridge or N. [S2–S4: proved, sign-off pending]; whether the non-collapse condition in the PIP theorem is necessary [open].

E. Limits (proved negative results)

- The greedy search from the start mode alone can fail: (i) an explicit design satisfying all other hypotheses gives $P(\text{search misses } S_0) \to 1$ while $\pi(S_0) \to 1$ (older move cap); (ii) on exact-Gram compound symmetry with $K < p \le n-1$, $\liminf p/K > 16/15$, the shipped search never scores S_0. Both concern that search route, not the full method. [proved, checked]
- Route-level obstructions: for compound symmetry (one factor) with $\liminf \log p/\log n > 1/6$ the no-addition event the deletion route needs fails (Prop. SHARP); for ≥ 2 factors the uniform subset information the current technique needs does not exist (OBS-k). [proved, checked]
- Sharpness: rate lower bound (A), specificity counterexample (A), threshold constant in C1.

4 Open questions and next steps

1. **Strong factors at $p \gg n$.** Experiment (paused, resumes with the project) running the full method on factor and equicorrelation designs with $p > K$, p/n up to 10. So far: the $n = 50$ batch is nearly complete (298 of 320 fits); $n = 75$–150 not yet run. No conclusion yet. The outcome decides whether the paper claims recovery or only screening plus the limits above for this class.
2. **Confirmation rerun** of the 2,400-fit study on the final frozen version, about 910 core-hours. Islands are unchanged, so $\hat{Z}$, island $\hat{Z}$, RB PIPs and MPM are expected bit-identical and the HPM score can only rise.
3. **Scaling programme.** Make the method feasible where sequential samplers break (p in the thousands to tens of thousands): many independent islands on a cluster, parallel block tables, on-demand Gram computation; merge islands by evidence weights $\sum_r \hat{Z}_r \hat{h}_r / \sum_r \hat{Z}_r$ (each island unbiased by S1), HPM from the union of scored models. Theory for the parallel method is part of the plan; *weight degeneracy of the island evidences at high p is open.*
4. **Remaining proofs:** start-mass rate; polynomial-size leakage sets $|Q| \le n^b$ in C2; removing the lattice condition (N).
5. **Longer term:** the same machinery (exact decomposable start law, island SMC with evidence merging, certified search output) for other huge discrete model spaces: graphical models, trees.

5 Questions for the advisor

1. Which result should be the paper's headline: posterior concentration with the matching lower bound, the reported-model theorem (C2), or design-free certified screening (C1)?
2. Should the parallel many-island method and its theory go in this paper or in a second paper?
3. Is the lattice condition (N) acceptable as a stated hypothesis, or should it be removed before submission?
4. Target journal (e.g. JRSS-B, Annals, JASA Theory & Methods), given the mix of method, simulation and theory?